Nirbhay Singh

Cloud & DevOps Architect | Multi-Cloud Platform Engineer

11+	50%+	30x	EUR 4M+	14+
Years Experience	Faster Deploys	CI/CD Speedup	Cloud Savings	Certifications

PROFESSIONAL SUMMARY

Cloud & DevOps Architect with 11+ years designing, automating, and operating production infrastructure across AWS, GCP, and Azure simultaneously. Expert in Terraform IaC, Kubernetes multi-cluster management (EKS, GKE, AKS), and GitOps-driven deployment workflows. Proven record of reducing deployment time by 50%+, accelerating CI/CD pipelines by 30x, and orchestrating 1,000+ container workloads. Delivered EUR 4M+ in cloud cost savings through FinOps governance at Bosch Europe. Currently architecting multi-cloud strategy and platform engineering for one of Europe's largest industrial conglomerates.

KEY ACHIEVEMENTS

- Architected multi-cloud platform strategy across GCP and AWS for Bosch Europe -- unified governance, networking, and security across 15+ business units
- Automated infrastructure with Terraform IaC (1,500+ resources), achieving 50%+ reduction in deployment time and near-zero configuration drift
- Accelerated CI/CD pipeline speed by 30x at Deloitte through parallel execution, caching strategies, and pipeline-as-code architecture redesign
- Orchestrated 1,000+ containers using AWS Batch, Spot Instances, and EKS with automated scaling, health monitoring, and self-healing
- Deployed and managed production Kubernetes clusters across GKE, EKS, AKS, and ECS simultaneously with GitOps (ArgoCD) workflows
- Implemented FinOps governance delivering EUR 4M+ in savings through tagging strategies, right-sizing, and reserved capacity planning

PROFESSIONAL EXPERIENCE

Cloud & AI Architect

Dec 2022 - Present

Robert Bosch (Bosch Polska) Warsaw, Poland

- Architected multi-cloud infrastructure strategy across GCP and AWS for Bosch Europe business units -- defined landing zones, networking, identity, and governance
- Led strategic partnership between Google Cloud, AWS, and Bosch Poland -- unified cloud tooling, established architecture review boards, and standardised deployment patterns
- Designed GKE and Cloud Run microservices platforms for enterprise AI workloads with autoscaling, service mesh, and zero-trust security
- Implemented FinOps at enterprise scale, delivering EUR 4M+ in savings through optimisation, tagging governance, and cost-aware architecture reviews
- Mentored 20+ senior developers and architects on multi-cloud strategy, IaC best practices, and Kubernetes operations across global teams

Senior DevOps Consultant

Jun 2021 - Nov 2022

McAfee

- Automated infrastructure provisioning with Terraform IaC across AWS, GCP, and Azure -- 50%+ reduction in deployment time, near-zero drift
- Deployed and managed production container platforms across GKE, EKS, AKS, and ECS with GitOps workflows and automated rollbacks
- Built comprehensive observability stack: Prometheus, Grafana, and ELK for metrics, logging, alerting, and distributed tracing
- Developed Python and Bash automation for infrastructure lifecycle management, secret rotation, and compliance scanning

Nirbhay Singh

Multi-Cloud | Kubernetes | Terraform
CI/CD | GitOps | Platform Engineering

CONTACT

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CORE SKILLS

CLOUD PLATFORMS

AWS, GCP, Azure, VMware, Multi-Cloud Architecture, Landing Zones

IAC & CONFIG

Terraform (expert), CloudFormation, Ansible, Puppet, HCL, Policy-as-Code

CONTAINERS

Kubernetes (EKS, GKE, AKS), Docker, Helm, Anthos, ECS, Istio

CI/CD & GITOPS

GitHub Actions, Jenkins, ArgoCD, Flux, GitOps, Spinnaker

OBSERVABILITY

Prometheus, Grafana, ELK Stack, Datadog, CloudWatch

SECURITY

IAM, Zero-Trust, OPA/Gatekeeper, Secret Mgmt, Compliance

LANGUAGES

Python, Bash, YAML, JSON, HCL, PowerShell, Go

EDUCATION

MTech (Master of Technology)

Birla Institute of Technology

LANGUAGES

English	Professional
Hindi	Native

Senior DevOps Engineer

Dec 2018 - Jun 2021

Deloitte

- Reduced AWS costs by $45,000 and accelerated CI/CD pipeline speed by 30x through architecture redesign, caching, and parallel execution
- Orchestrated 1,000+ containers using AWS Batch, Spot Instances, and EKS for distributed batch processing and analytics
- Implemented multi-cloud security controls and networking spanning GCP, AWS, Azure, and OCI with unified IAM and audit logging
- Built data platform infrastructure on EMR, Redshift, and BigQuery with automated ETL pipelines and data governance

Senior Cloud Infrastructure Engineer

Nov 2015 - Dec 2018

PTC

- Designed secure, fault-tolerant AWS architectures (EC2, RDS, S3, VPC) with multi-AZ deployments and automated failover
- Built disaster recovery strategies for hybrid cloud and on-premise environments with RTO/RPO guarantees
- Managed 11,000+ server Windows infrastructure with Active Directory, Group Policy, and operations automation
- Led large-scale cloud migration planning and execution using PowerShell, Ansible, and custom migration tooling

System Engineer

Mar 2013 - Jul 2014

IBM

- Infrastructure operations, migrations, and P1/P2 incident response for enterprise production environments

OPEN-SOURCE PROJECTS

ShieldIaC -- AI-Powered IaC Security Scanner

>> github.com/nirbhays/shieldiac

Catches security misconfigurations in Terraform and CloudFormation before they reach production. 100+ rules, 9 compliance frameworks (CIS, SOC2, HIPAA, PCI-DSS, NIST), AI-powered fix suggestions via GPT-4.1.

STACK: Python, Terraform, CloudFormation, GPT-4.1, GitHub Actions

InfraCents -- Terraform Cost Estimates on Every PR

>> github.com/nirbhays/infracents

GitHub App posting real-time AWS+GCP cost estimates on pull requests before merge. Zero config, automatic Terraform parsing, live pricing APIs.

STACK: Python, Next.js, Terraform, GitHub App, AWS/GCP Pricing APIs

TokenMeter -- LLM Cost Intelligence Layer

>> github.com/nirbhays/tokenmeter

Tracks token usage, cost, and latency across OpenAI, Anthropic, Google. Smart routing saves up to 60% on AI spend. <5ms overhead.

STACK: Python, OpenAI, Anthropic, Google AI, Smart Routing, PyPI

CERTIFICATIONS & CREDENTIALS

- **AWS** Solutions Architect - Professional (Feb 2020)
- **Google** Professional Cloud Architect (Valid Jun 2026)
- **Google** Associate Cloud Engineer (Aug 2023)
- **HashiCorp** Terraform Associate
- **Google** Generative AI Leader (Valid Aug 2028)
- **AWS** Solutions Architect - Associate (Jan 2021)
- **Google** Professional Cloud DevOps Engineer (Valid Dec 2027)
- **CKAD** - Certified Kubernetes Application Developer (Valid Sep 2026)
- **Google** Professional ML Engineer (Valid Dec 2027)
- **MCSA** - Microsoft Certified Solutions Associate (Apr 2015)